

Signature:

I hereby confirm that all the pages of my exam paper
have been stapled to this cover sheet.

End-of-Module Exam: 2026 Spring Semester

Zürcher Hochschule

für Angewandte Wissenschaften

zh
aw School of
Management and Law

Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	26.08.2026 / 14:00 – 15:00
Duration:	60 Minutes
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

End-of-Module Exam: 2026 Spring Semester
Module Code: w.BA.XX.3DSI-WIN

Zürcher Hochschule

für Angewandte Wissenschaften

zh
aw School of
Management and Law

Name: Vorname:

Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		<i>(Grade on cover sheet)</i>	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	30 Points	hitz
	Total	30 Points	

End-of-Module Exam: 2026 Spring Semester

Module Code: w.BA.XX.3DSI-WIN

Zürcher Hochschule

für Angewandte Wissenschaften

zh
aw School of
Management and Law

Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.

ONLY tick ONE option per question in this table.

Scoring: Each correctly answered question is worth **1 point**.

Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	☐ A ☐ B ☐ C ☐ D	16.	☐ A ☐ B ☐ C ☐ D
2.	☐ A ☐ B ☐ C ☐ D	17.	☐ A ☐ B ☐ C ☐ D
3.	☐ A ☐ B ☐ C ☐ D	18.	☐ A ☐ B ☐ C ☐ D
4.	☐ A ☐ B ☐ C ☐ D	19.	☐ A ☐ B ☐ C ☐ D
5.	☐ A ☐ B ☐ C ☐ D	20.	☐ A ☐ B ☐ C ☐ D
6.	☐ A ☐ B ☐ C ☐ D	21.	☐ A ☐ B ☐ C ☐ D
7.	☐ A ☐ B ☐ C ☐ D	22.	☐ A ☐ B ☐ C ☐ D
8.	☐ A ☐ B ☐ C ☐ D	23.	☐ A ☐ B ☐ C ☐ D
9.	☐ A ☐ B ☐ C ☐ D	24.	☐ A ☐ B ☐ C ☐ D
10.	☐ A ☐ B ☐ C ☐ D	25.	☐ A ☐ B ☐ C ☐ D
11.	☐ A ☐ B ☐ C ☐ D	26.	☐ A ☐ B ☐ C ☐ D
12.	☐ A ☐ B ☐ C ☐ D	27.	☐ A ☐ B ☐ C ☐ D
13.	☐ A ☐ B ☐ C ☐ D	28.	☐ A ☐ B ☐ C ☐ D
14.	☐ A ☐ B ☐ C ☐ D	29.	☐ A ☐ B ☐ C ☐ D
15.	☐ A ☐ B ☐ C ☐ D	30.	☐ A ☐ B ☐ C ☐ D

End-of-Module Exam: 2026 Spring Semester
Module Code: w.BA.XX.3DSI-WIN

Zürcher Hochschule
für Angewandte Wissenschaften

zh
aw School of
Management and Law

MC Data Science Introduction (30 Points)

1. Which of the following best distinguishes the Viable System Model (VSM) from System Dynamics (SD)? (1)
 - A. VSM is a recursive management structure for viability, while SD is a simulation method for behaviour over time.
 - B. VSM simulates behaviour over time, while SD is essentially a static organisational chart.
 - C. They are two names for exactly the same method.
 - D. Neither VSM nor SD makes any use of feedback.
2. A firm in a volatile market adds data sources and flexible processes while also focusing its product portfolio. In Ashby's terms this combines... (1)
 - A. neither amplification nor attenuation.
 - B. amplification (more own variety) and attenuation (less environmental variety).
 - C. a deliberate reduction of requisite variety.
 - D. only attenuation.
3. Which single statement about the distinction of Data Science is correct? (1)
 - A. Analytics is broader than the whole of Data Science.
 - B. Data Science deliberately excludes domain knowledge.
 - C. Possessing big data is the same as doing data science.
 - D. A dataset is 'big' when its size becomes central to the problem being solved.
4. Which statement about real-time adaptive models is NOT correct? (1)
 - A. They rely on continuous feedback loops.
 - B. They can incorporate reinforcement learning.
 - C. Once deployed they never need updating because they are 'adaptive' by name only.
 - D. They are well suited to fast-changing (VUCA) environments.
5. Which statement about overfitting is NOT correct? (1)
 - A. Overfitting means the model is too simple to capture patterns.
 - B. Overfitting performs well on training data but poorly on unseen data.
 - C. Data leakage can produce an overfitting-like over-optimism.
 - D. Regularization and more data can reduce overfitting.

6. Which statement about visualizing results is correct? (1)
- A. Visual literacy is never required to read a chart.
 - B. A truncated axis can exaggerate small differences, violating the true-and-fair view.
 - C. Visualizations always inform objectively.
 - D. A well-designed chart can never mislead.
7. A smart city's sensor data is high-volume, noisy, heterogeneous and temporally dependent. This best illustrates... (1)
- A. a perfectly balanced dataset.
 - B. underfitting.
 - C. a simple, IID dataset.
 - D. data complexity.
8. Which single statement about class weighting is correct? (1)
- A. It is just another name for SMOTE.
 - B. It gives the minority class more importance in the loss without changing the dataset.
 - C. It physically deletes majority-class rows.
 - D. It duplicates minority-class rows.
9. Which statement about target-oriented action is NOT correct? (1)
- A. The goal drives data and method choices.
 - B. Measurable targets help track progress.
 - C. The cycle culminates in a concrete action.
 - D. Projects succeed best by starting from available data and ignoring the goal.
10. Which statement about explainability is NOT correct? (1)
- A. Traceability makes system logic and outcomes auditable.
 - B. Explainability is irrelevant in high-stakes decisions.
 - C. Explainability improves user trust in AI decisions.
 - D. Large foundation models are hard to interpret due to huge, opaque parameter spaces.
11. In control terms, Ashby's Law requires that... (1)
- A. the environment be ignored.
 - B. the controller be simpler than the system it steers.
 - C. the controller have at least as much variety as the system it steers.
 - D. more data alone guarantees control.
12. A corporation whose divisions largely self-regulate, with the centre intervening only when instability arises, mirrors... (1)
- A. an ROC curve.
 - B. the Viable System Model.
 - C. a data swamp.
 - D. the SMOTE algorithm.

13. Which single statement about handling class imbalance is correct? (1)
- A. Oversampling deletes majority-class examples.
 - B. Imbalance is best ignored.
 - C. Accuracy is the most reliable metric under strong imbalance.
 - D. SMOTE creates synthetic minority samples by interpolation.
14. Which statement about uncertainty is NOT correct? (1)
- A. Uncertainty implies analysis is fundamentally impossible.
 - B. Scenario analysis helps cope with it.
 - C. Outcomes are unknown despite available data.
 - D. It is distinct from ambiguity.
15. Which statement about unsupervised learning is NOT correct? (1)
- A. It predicts predefined output labels.
 - B. K-means is an unsupervised method.
 - C. It is well suited to exploration and pattern finding.
 - D. It discovers groupings without prior knowledge of outputs.
16. Ashby's Law of Requisite Variety states that... (1)
- A. data volume alone guarantees control.
 - B. environmental complexity is irrelevant to survival.
 - C. only variety can absorb variety - a controller must match the system's variety.
 - D. simpler systems are always better.
17. The bullwhip effect in supply chains is best analysed with... (1)
- A. the SMOTE algorithm.
 - B. System Dynamics (feedback loops plus delays).
 - C. a confusion matrix.
 - D. zero-shot learning.
18. Which single statement about smart factories is correct? (1)
- A. MES link production machinery (physical) with ERP systems (cyber) for real-time control.
 - B. They avoid analytics entirely.
 - C. They run on fixed schedules with no data at all.
 - D. They are purely offline reporting systems.
19. Which statement about model evaluation is NOT correct? (1)
- A. The confusion matrix underlies many classification metrics.
 - B. The ROC curve shows the trade-off between hit rate and false alarms.
 - C. Recall measures the proportion of predicted positives that are correct.
 - D. Recall matters most when false negatives are costly.

20. Which statement about unsupervised algorithms is NOT correct? (1)
- A. K-means requires labeled training data.
 - B. K-medoids uses a real data object as the centre.
 - C. Fuzzy C-means allows partial membership in several clusters.
 - D. K-means assigns each point to the nearest centroid.
21. The target-centric approach is best described as... (1)
- A. a linear, one-shot analysis.
 - B. a structured, iterative method that continuously refines insights as new data emerges.
 - C. a pure visualization tool.
 - D. a data-first exploration with no defined goal.
22. Identifying influential nodes and tightly connected groups in a social graph is an example of... (1)
- A. zero-shot learning.
 - B. clustering of tabular features only.
 - C. linear regression.
 - D. network analysis.
23. Which single statement about Big Data / Data Science is actually TRUE (not a myth)? (1)
- A. About 95% of big data is unstructured (text, audio, images).
 - B. More data is always better, regardless of quality.
 - C. Data lakes are inherently valuable even without metadata.
 - D. Merging sources automatically yields a single, complete 'Golden Record'.
24. Which statement about EDA is NOT correct? (1)
- A. EDA helps spot data-quality issues early.
 - B. It examines distributions and outliers.
 - C. It should be skipped to save time on large datasets.
 - D. It informs cleaning and feature engineering.
25. Which statement about Ashby's Law is NOT correct (i.e., is a misinterpretation)? (1)
- A. Reducing environmental variety (attenuation) is as valid as amplifying one's own.
 - B. The law is a guiding principle, not an exact formula.
 - C. Variety means distinguishable states, not data volume.
 - D. The only option is to keep adding internal complexity.
26. VUCA stands for... (1)
- A. Volatility, Uncertainty, Complexity, Ambiguity.
 - B. Variety, Uniqueness, Consistency, Accuracy.
 - C. Velocity, Utility, Clarity, Agility.
 - D. Volume, Uncertainty, Complexity, Analytics.

27. Eager learners typically have... (1)
- A. no model at all.
 - B. low training time and fast prediction.
 - C. high training time but fast prediction.
 - D. instance-based storage of all training data.
28. Which single statement about data bias is correct? (1)
- A. Bias is only ever a purely technical flaw.
 - B. Bias often begins with humans but becomes more dangerous when automated and scaled by AI.
 - C. Anonymized data carries zero risk of re-identification.
 - D. A neutral appearance proves the absence of bias.
29. Which statement about VSM early warning is NOT correct? (1)
- A. Anti-oscillatory loops dampen emerging instability.
 - B. Early detection enables timely intervention.
 - C. Removing feedback improves early warning.
 - D. SPC and anomaly detection aid early warning.
30. Which statement about interdisciplinarity is NOT correct? (1)
- A. It combines math/statistics, computer science/IT, domain knowledge and social science.
 - B. Cao's 'X-Intelligence' fuses many sources of intelligence.
 - C. Ignoring domain knowledge risks 'solitary analysis'.
 - D. Social science is irrelevant to data science.